

# ROBIN INHO KEE

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## Education

### University of Michigan

*Doctor of Philosophy in Robotics*

**Aug 2025 – Present**

*Ann Arbor, MI, USA*

- **Advisors:** Prof. Vasileios Tzoumas and Prof. Necmiye Ozay
- *Draper Scholars Program* – fully funded Ph.D. fellowship in autonomy (**Draper advisor:** Dr. Begum Cannataro)

### University of Michigan

*Master of Science in Mechanical Engineering*

**Aug 2023 – Dec 2024**

*Ann Arbor, MI, USA*

### Yonsei University

*Bachelor of Science in Mechanical Engineering*

**Mar 2016 – Feb 2022**

*Seoul, South Korea*

- *Mandatory military service: Republic of Korea Air Force, Oct 2020 – Jul 2022*

## Research Interests

Multi-Robot Autonomy, Communication-Aware Coordination, Learning-Enabled Control, Scalable Decision-Making

## Publications

\* *Equal contribution*

### Journals

- [J2] “Disentangled Prototypical Convolutional Network for Few-Shot Learning in In-Vehicle Noise Classification.”  
**R. I. Kee**, D. Nam, S.-J. Buu, and S.-B. Cho. *IEEE Access*, 2024. [\[Paper\]](#)
- [J1] “Measurement of ankle joint movements using IMUs during running.”  
B. Kim, S. Hong, I. Oh, Y. Lee, **I. Kee**, and S. Lee. *Sensors*, 2021. [\[Paper\]](#)

### Conferences

- [C5] “Time Shift Governor-Guided MPC with Collision Cone CBFs for Safe Adaptive Cruise Control in Dynamic Environments.”  
**R. I. Kee**\*, T. Kim\*, A. Girard, and I. Kolmanovsky. *IEEE CCTA*, 2025. [\[Paper\]](#)
- [C4] “Learning to Refine Input Constrained Control Barrier Functions via Uncertainty-Aware Online Parameter Adaptation.”  
T. Kim, **R. I. Kee**, and D. Panagou. *IEEE ICRA*, 2025. [\[Paper\]](#)
- [C3] “Constrained Control for Autonomous Spacecraft Rendezvous: Learning-Based Time Shift Governor.”  
T. Kim\*, **R. I. Kee**\*, I. Kolmanovsky, and A. Girard. *AIAA SciTech Forum*, 2025. [\[Paper\]](#)
- [C2] “Dynamic Prototype-guided Memory Replay for In-Vehicle Noise Classification.”  
D. Nam, **I. Kee**, S.-J. Bu, and S.-B. Cho. *Korea Data Mining Society*, 2023. **SAS Student Paper Award**. [\[Project Page\]](#)
- [C1] “An Intelligent Lock-Out Tag-Out System for Monitoring and Control of the Locked Device.”  
W. Jo, S. Hwang, **I. Kee**, I. Lee, and S. Lee. *IEEE IEEM*, 2019. [\[Project Page\]](#)

### Manuscripts Under Review / Preprints

- [P4] “AC-DC: Adaptive Communication for Scalable Dynamic Average Consensus in Multi-Robot Ergodic Search.”  
**R. I. Kee**, B. Cannataro, and V. Tzoumas. *IEEE ICRA*, 2027 (submitted). [\[arXiv\]](#)
- [P3] “Learning to Adapt Control Barrier Functions Under Epistemic and Aleatoric Uncertainty.”  
T. Kim, **R. I. Kee**, and D. Panagou. *IEEE T-RO* (under review). [\[arXiv\]](#)
- [P2] “Toward Daily-Life Sarcopenia Identification via Real-Time Gait Assessment Using a Wearable Hip Assistive Robot.”  
S. Kim\*, S. Kim\*, J. Kang, **R. I. Kee**, K. R. Mun, Y. Lee, K. Kim, Y. Cha, and J. Lee. *IEEE RA-L* (under review). [\[Project Page\]](#)
- [P1] “Vision-Ultrasound Robotic System based on Deep Learning for Gas and Arc Hazard Detection in Manufacturing.”  
J.-H. Lee, D. Nam, **R. I. Kee**, Y. Kim, and S.-J. Bu. *arXiv preprint*, 2025. [\[arXiv\]](#)

## Research Experience

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### Graduate Research Assistant, University of Michigan

Aug 2025 – Present

*Intelligent Robotics and Autonomy Lab (PI: Prof. Vasileios Tzoumas)*

*Ann Arbor, MI, USA*

- Conceived, designed, and implemented AC-DC, a peer-to-peer dynamic consensus framework that jointly adapts communication partners, timing, and consensus-state blocks under range, rate, and interference constraints
- Evaluated AC-DC across 12 settings with 20 paired trials each; reduced mean paired AUC by 29.5% and 24.5% versus ADMM-DAC and PP-ACDC, while those baselines used  $5.8\times$  and  $11.5\times$  as much modeled payload [P4]

### Graduate Research Assistant, University of Michigan

May 2024 – Aug 2025

*Distributed Autonomous Systems and Control Lab (PI: Prof. Dimitra Panagou)*

*Ann Arbor, MI, USA*

- Implemented and integrated the OA-CBF research pipeline by adapting control and learning components and extending the codebase with graph-attention and uncertainty-aware screening for online CBF-parameter selection
- Conducted the complete evaluation across five nonlinear systems; achieved 0% safety failures and 100% goal-reaching in three 100-trial benchmarks, with 39–65% shorter reach times than the safe fixed-parameter baseline [C4/P3]

### Graduate Research Assistant, University of Michigan

Jan 2024 – Feb 2025

*VODCA Lab (PIs: Profs. Anouck Girard and Ilya Kolmanovsky)*

*Ann Arbor, MI, USA*

- Developed the LSTM prediction module for a hybrid Time Shift Governor with online verification and a bisection fallback; reduced average time-shift computation by 84% in LEO and 70% in Molniya-orbit spacecraft-rendezvous simulations [C3]
- Designed and implemented a TSG-guided MPC-CBF controller for adaptive cruise control in dynamic curved-road scenarios; achieved 100% success over 50 randomized trials versus 82% for baseline MPC-CBF [C5]

### Research Intern, Korea Institute of Science and Technology

Jul 2022 – Jul 2023

*Assistive and Interactive Robotics Lab (PI: Dr. Jongwon Lee)*

*Seoul, South Korea*

- Built the initial end-to-end sensing and learning pipeline for a 4-DOF wearable hip robot, including multimodal sensor fusion, motion-capture synchronization, preprocessing, and foot-trajectory and gait-parameter estimation
- Established the experimental pipeline and led initial human-subject studies; the resulting 56-participant, 51,610-stride study achieved 6.69 cm stride-length RMSE and 90.17% stride-wise sarcopenia-identification accuracy [P2]

## Technical Skills

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**Programming & Tools:** Python, C/C++, MATLAB, PyTorch, Linux, Git

**Robotics & Methods:** ROS/ROS 2, MPC, CBFs, Online Learning, Multi-Robot Coordination

## Selected Awards and Honors

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**Draper Scholars Program (Full Ph.D. Fellowship)**, Charles Stark Draper Laboratory, 2025–Present

**SAS Student Paper Award**, Korea Data Mining Society, 2023

**1st Place, International S.M.A.R.T Startup Competition**, Innovative Technology and Energy Center, 2020

**Science and Engineering Undergraduate Internship Program Scholarship**, Yonsei University, 2020

**Academic Highest Honors**, Yonsei University, 2019, 2020

**2nd Prize, Robot Open Source Lab**, Samsung Open Source Conference, 2019

**Academic Excellence Scholarship**, Yonsei University, 2018, 2019

## Teaching & Academic Service

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**Graduate Student Instructor:** AEROSP 740, Online Learning for Single- and Multi-Agent Control

Fall 2026

- Developed lab materials and code for quadrotor simulation and Crazyflie hardware experiments; held weekly office hours and supported simulation and hardware assignments

**Reviewer:** ICRA 2027; CCTA 2026    **Conference Volunteer:** L4DC 2025

## Leadership

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### Roboin Robotics Club

Jun 2018 – Aug 2023

*President (2019); Advisory Committee (2020–2023)*

*Yonsei University, Seoul, South Korea*

- Selected projects: Autonomous Fire-Extinguishing Robot [Project Page]; Theo-Jansen Legged Robot [Project Page]